

MASCOT
GOES
HERE

KID INSTRUCTIONS

Algorithms work because the world has PATTERNS — repeating shapes, growing numbers, hidden rules. Today, you become a pattern detective: find the rule, predict what's next!

Pattern

Pattern — A rule that tells you what comes next in a sequence. Find the rule = predict the future!

Example: Your weekly schedule has a pattern — school, school, school, school, school, weekend! Once you know the rule, you can predict any day.

How It Works — Three Types of Patterns

There are three main types of patterns. Once you know them, you can spot patterns almost anywhere!

1. REPEATING

the same thing happens over and over

Example:       

Rule: The loop is "square, circle." → Next: square

2. GROWING

each step gets bigger by a rule

Example: 2, 4, 6, 8, ____
Rule: The rule is "+2 each time." → Next: 10

3. RULE-BASED

every step follows a hidden instruction

Example: 1, 4, 9, 16, ____
Rule: Each number is its position squared. → Next: 25

Detective Trick: Try to find the rule in just 2-3 items before you guess what's next!

Basic Version

Find the Next One — For each pattern, figure out the rule and predict the next item!

Pattern 1



What's the rule?

→

What comes next?

Pattern 2

5, 10, 15, 20, ____

What's the rule?

→

What comes next?

Pattern 3

A, B, C, A, B, C, A, ____

What's the rule?

→

What comes next?

Pattern 4

2, 4, 8, 16, ____

What's the rule?

→

What comes next?

Challenge Version

PART A — Spot the Hidden Rule

These are trickier! For each pattern, identify the type, find the rule, and predict what comes next.

Pattern A1

1, 1, 2, 3, 5, 8, ____

Type:

☐ Repeating ☐ Growing ☐ Rule-based

Rule:

→

Next:

Pattern A2

100, 90, 81, 73, 66, ____

Type:

☐ Repeating ☐ Growing ☐ Rule-based

Rule:

→

Next:

Pattern A3

M, T, W, T, F, S, ____

Type:

☐ Repeating ☐ Growing ☐ Rule-based

Rule:

→

Next:

PART B — Write Your Own Pattern

Now it's your turn! Make up your own pattern. Write the first 5 items, then keep the rule a secret so others have to figure it out.

My pattern type:

☐ Repeating ☐ Growing ☐ Rule-based

First 5 items of my pattern:

The hidden rule:

(don't share with the person trying to guess!)

→

→

What comes next?

Self-Check

- ☐ I found the rule and next item for all 4 Basic patterns
- ☐ I named the type (repeating, growing, rule-based) for all 3 Challenge patterns
- ☐ I invented my own pattern with a hidden rule
- ☐ I can explain the 3 pattern types in my own words

Reflection Box

Real programmers use pattern recognition to FIND BUGS, predict user behavior, and even teach computers to recognize cats in photos. Can you think of one pattern your phone or favorite app has noticed about YOU?

Bonus: Pattern Hunter

Go on a pattern hunt around your home! Find 4 real patterns in everyday things — wallpaper, tiles, music, the calendar, even how often you eat meals. Patterns are EVERYWHERE once you start looking.

Pattern #1

Where I found it:

What's the pattern?

Type: ☐ Repeating ☐ Growing ☐ Rule-based

Pattern #2

Where I found it:

What's the pattern?

Type: ☐ Repeating ☐ Growing ☐ Rule-based

Pattern #3

Where I found it:

What's the pattern?

Type: ☐ Repeating ☐ Growing ☐ Rule-based

Pattern #4

Where I found it:

What's the pattern?

Type: ☐ Repeating ☐ Growing ☐ Rule-based

Which was the most surprising pattern you found? Why?

TEACHER / PARENT NOTE

Pattern recognition is the underlying skill behind almost every algorithm — from sorting (Activities 8-9) to debugging (Activity 7), and beyond that into search engines, recommendation systems, and machine learning. The 3 pattern types kids learn here map directly to real CS structures: repeating ↔ loops, growing ↔ recursion, rule-based ↔ functions. Discussion prompt: Ask your child to predict a pattern about your family's week — when does laundry get done? when is takeout night? — and see if they can predict the next "event" based on the pattern.